

Finding Inverses with Domain Restrictions

Answer Key

Precalculus Advanced Function Transformations and Composition

Quick Answers

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|------------|-------------|----------------|-------|
| 1. $x, 5$ | 4. $x, 10$ | 7. $11, 5$ | 10. — |
| 2. $13, 3$ | 5. $-6, 11$ | 8. $-1, 9$ | |
| 3. $7, 12$ | 6. $-2, 5$ | 9. $-4, -1, 5$ | |
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Curriculum

Level: Precalculus Advanced Function Transformations and Composition

HSF-BF.B.4 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

HSF-IF.A–HSF-IF.C – *problems 2, 6, 9*

Difficulty 1–5 of 5 across 20 tagged checks.

Common wrong answers

8. *If they got 77: used the claimed rule, squaring the input instead of taking its square root*

1. $f(x) = 3x - 7$. **Find f^{-1} and $f^{-1}(8)$.**

Step 1. A non-constant linear function is one-to-one everywhere, so no restriction is needed and the inverse is defined on all real numbers.

Step 2. Write $y = 3x - 7$, swap: $x = 3y - 7$, then solve: $y = \frac{x+7}{3}$.

$$f^{-1}(x) = \frac{x+7}{3}$$

Step 3. $f^{-1}(8) = \frac{8+7}{3} = \frac{15}{3}$, so

$$f^{-1}(8) = 5$$

Check by composition: $f(f^{-1}(x)) = 3 \cdot \frac{x+7}{3} - 7 = x$, so the pair is genuinely inverse.

2. $f(x) = \sqrt{x-4}$ on $x \geq 4$. **Find f^{-1} , its domain, and $f^{-1}(3)$.**

Step 1. A square root is increasing on its whole domain, so f is already one-to-one. Its outputs are never negative, so the range of f is $y \geq 0$ — and that range becomes the domain of the inverse.

Step 2. $y = \sqrt{x-4}$; swap and square: $x^2 = y - 4$, so $y = x^2 + 4$. The squaring step is exactly why a domain must be attached: $x^2 + 4$ is defined for every real x , but only $x \geq 0$ can be an output of f .

$$f^{-1}(x) = x^2 + 4, \ x \geq 0$$

Step 3. $f^{-1}(3) = 3^2 + 4$, so

$$f^{-1}(3) = 13$$

Step 4. Check the other way: $f(13) = \sqrt{13-4} = \sqrt{9}$, so
— the round trip returns the input.

$$f(13) = 3$$

3. $f(x) = x^2 - 6x + 5$ on $x \geq 3$. Find f^{-1} , its domain, and $f^{-1}(12)$.

Step 1. A parabola is not one-to-one, so the restriction is doing real work: complete the square to find the vertex and confirm the restriction starts there. $f(x) = (x-3)^2 - 4$, vertex at $(3, -4)$, and $x \geq 3$ keeps the right-hand branch.

Step 2. $y = (x-3)^2 - 4$; swap and solve: $(y-3)^2 = x+4$, so $y-3 = \pm\sqrt{x+4}$. On the kept branch $y \geq 3$, so the sign is positive.

$$f^{-1}(x) = 3 + \sqrt{x+4}, \quad x \geq -4$$

Step 3. $f^{-1}(12) = 3 + \sqrt{12+4} = 3+4$, so

$$f^{-1}(12) = 7$$

Step 4. Check: $f(7) = 49 - 42 + 5$, so

$$f(7) = 12$$

4. $g(x) = \frac{2x+1}{x-3}$. Verify g^{-1} , exclude a value, compute $g^{-1}(3)$.

Step 1. A composition check is the definitive test, so substitute the claimed inverse into g and simplify.

Step 2. With $u = \frac{3x+1}{x-2}$:

$$2u+1 = \frac{6x+2+x-2}{x-2} = \frac{7x}{x-2}, \quad u-3 = \frac{3x+1-3x+6}{x-2} = \frac{7}{x-2}.$$

Step 3. Dividing, the $(x-2)$ denominators cancel and the 7s cancel: $g(u) = \frac{7x}{7} = x$. The claimed formula is the inverse.

Step 4. The excluded value is $x = 2$. It comes from the horizontal asymptote of g : $g(x) \rightarrow 2$ as $x \rightarrow \pm\infty$ but never equals 2, so 2 is missing from the range of g and therefore from the domain of g^{-1} .

Step 5. $g^{-1}(3) = \frac{9+1}{3-2}$, so

$$g^{-1}(3) = 10$$

5. $f(x) = (x+2)^2 - 5$ on $x \leq -2$. Find f^{-1} , its domain, and $f^{-1}(11)$.

Step 1. The vertex is at $(-2, -5)$ and the restriction keeps the *left* branch, so every input of f is at most -2 — which means every output of f^{-1} must be at most -2 . That is what will decide the sign.

Step 2. $y = (x+2)^2 - 5$; swap and solve: $(y+2)^2 = x+5$, so $y = -2 \pm \sqrt{x+5}$. The plus sign would give outputs above -2 , so take the minus sign.

$$f^{-1}(x) = -2 - \sqrt{x+5}, \quad x \geq -5$$

Step 3. $f^{-1}(11) = -2 - \sqrt{16} = -2-4$, so

$$f^{-1}(11) = -6$$

Step 4. Check: $f(-6) = (-4)^2 - 5 = 16-5$, so

$$f(-6) = 11$$

Common wrong answer: 2 — the positive branch was taken out of habit. The kept branch, not the algebra, chooses the sign.

6. $f(x) = 9 - x^2$ on $x \leq 0$. Find f^{-1} , its domain and range, and $f^{-1}(5)$.

Step 1. This parabola opens downward with vertex at $(0, 9)$, and the restriction keeps the left half, on which f is increasing. Its outputs there run up to 9, so the range of f is $y \leq 9$.

Step 2. $y = 9 - x^2$; swap and solve: $y^2 = 9 - x$, so $y = \pm\sqrt{9 - x}$. Outputs of f^{-1} must be at most 0, so take the negative root.

$$f^{-1}(x) = -\sqrt{9 - x}$$

Step 3. Domain of f^{-1} : $x \leq 9$ (the range of f). Range of f^{-1} : $y \leq 0$ (the restricted domain of f). The two swap, always.

Step 4. $f^{-1}(5) = -\sqrt{9 - 5} = -\sqrt{4}$, so

$$f^{-1}(5) = -2$$

Step 5. Check: $f(-2) = 9 - 4$, so

$$f(-2) = 5$$

7. $f(x) = \sqrt{2x - 6} + 1$ on $x \geq 3$. Find f^{-1} , its domain, and $f^{-1}(5)$.

Step 1. Peel the operations off in reverse order. The outermost step of f is “add 1”, and the square root is never negative, so the smallest output is 1: the range of f is $y \geq 1$.

Step 2. $y = \sqrt{2x - 6} + 1$; swap and isolate the radical: $x - 1 = \sqrt{2y - 6}$. Square:

$$(x - 1)^2 = 2y - 6, \text{ so } y = \frac{(x - 1)^2 + 6}{2}.$$

$$f^{-1}(x) = \frac{(x - 1)^2 + 6}{2}, \quad x \geq 1$$

Step 3. $f^{-1}(5) = \frac{16 + 6}{2} = \frac{22}{2}$, so

$$f^{-1}(5) = 11$$

Step 4. Check: $f(11) = \sqrt{22 - 6} + 1 = 4 + 1$, so

$$f(11) = 5$$

Why the domain matters: $\frac{(x - 1)^2 + 6}{2}$ is defined for every real x , but only $x \geq 1$ comes from an output of f . Without the restriction the formula is a different function.

8. Priya’s claim about $f(x) = (x + 4)^2$ on $x \geq -4$.

Step 1. Priya reversed the shift but not the squaring: her rule squares the input, when undoing a square requires a square *root*. Test at $x = 9$ to see it fail.

Step 2. Correct method: $y = (x + 4)^2$; swap and solve: $y + 4 = \pm\sqrt{x}$. The restriction keeps $x \geq -4$, so outputs of f^{-1} satisfy $y \geq -4$ and the positive root is the right one: $y = \sqrt{x} - 4$.

Step 3. $f^{-1}(9) = \sqrt{9} - 4 = 3 - 4$, so

$$f^{-1}(9) = -1$$

Step 4. Check: $f(-1) = (-1 + 4)^2 = 3^2$, so

$$f(-1) = 9$$

Step 5. The correct formula, with its domain (the range of f , which is $x \geq 0$):

$$f^{-1}(x) = \sqrt{x} - 4, \quad x \geq 0$$

Common wrong answer: 77 — Priya’s rule at $x = 9$. She undid the “+4” correctly but squared where she should have taken a root.

9. $f(x) = x^2 + 8x + 12$: restrict, invert, evaluate.

Step 1. A parabola is one-to-one on either side of its vertex and on no larger interval, so the largest interval of the form $x \geq c$ is the one that starts exactly at the vertex.

Step 2. The axis of symmetry solves $2x + 8 = 0$, giving $x = -4$, so

$$x \geq -4$$

and $c = -4$. Completing the square, $f(x) = (x + 4)^2 - 4$.

Step 3. $y = (x + 4)^2 - 4$; swap and solve: $(y + 4)^2 = x + 4$, so $y = -4 \pm \sqrt{x + 4}$. The kept branch has $y \geq -4$, so take the plus sign.

$$f^{-1}(x) = -4 + \sqrt{x + 4}, \quad x \geq -4$$

Step 4. $f^{-1}(5) = -4 + \sqrt{9} = -4 + 3$, so

$$f^{-1}(5) = -1$$

Step 5. Check: $f(-1) = 1 - 8 + 12$, so

$$f(-1) = 5$$

10. Why one-to-one is required. (Manual review — open reasoning.)

Model answer. An inverse function must send each output of f back to *one* input. If two different inputs share an output, there is no single value to send back, so no function can do it.

Step 1. With $f(x) = x^2 - 4$ on all real numbers, $f(3) = 9 - 4 = 5$ and $f(-3) = 9 - 4 = 5$. Two different inputs, one output.

Step 2. So $f^{-1}(5)$ would have to be both 3 and -3 . A function assigns exactly one output to each input, so no such function exists: f has no inverse function on all of the real line.

 $f^{-1}(5)$ is not single-valued

Step 3. The repair is to restrict the domain to one side of the vertex. Taking $x \geq 0$ keeps only the right branch, on which f is increasing and therefore one-to-one, and there $f^{-1}(x) = \sqrt{x + 4}$.

Step 4. The domain of that inverse is $x \geq -4$ (the range of the restricted f) and its range is $y \geq 0$ (the restricted domain). Choosing $x \leq 0$ instead is equally valid and gives $f^{-1}(x) = -\sqrt{x + 4}$ with range $y \leq 0$ — which is exactly why the branch must be stated.

Grading note: full credit needs a concrete pair of inputs with the same output, the statement that a function cannot return two values, a named restriction, and the domain and range of the resulting inverse. A student who only says “it fails the horizontal line test” has restated the condition rather than explained it.